

CSE / EEE / ETE 499A (Section 1)

Project Impact and Constraints (CO4)

Project Title:

ShopMate AI — A Catalog-Grounded, Omnichannel AI Sales & Support Agent for Facebook, Instagram, WhatsApp, and Website Commerce

Submitted To

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Sustainability considerations for ShopMate AI are assessed across the three interdependent pillars economic, environmental, and social alongside the legal/standards constraints that shape the system's design. As a cloud-hosted software SaaS product rather than a physical device, several of the template's hardware-oriented prompts (e.g., renewable-energy tax credits, GHG-emitting manufacturing processes) do not directly apply; those points are noted as not applicable and replaced with the closest analogous software-industry consideration (e.g., cloud energy consumption in place of manufacturing emissions).

1. Economics (Cost) Impact

1.1 Effect on Final Product Cost / Pricing

ShopMate AI's per-tenant cost is dominated by metered LLM inference rather than fixed manufacturing cost, so its pricing is directly exposed to two economic variables:

- **LLM API pricing volatility** the project's tiered-model-routing design (CO2, Section 6.6) is itself a cost-mitigation response to this: cheap/fast models absorb classification-heavy steps so that a provider price change on the frontier model used only for final generation does not proportionally move the whole system's cost base.
- **Cloud infrastructure cost** using managed, pay-as-you-go cloud services (Section 4.6 of the CO1 proposal) instead of provisioned on-premise servers keeps fixed cost near zero during the pre-revenue capstone phase, but means unit economics only become favourable once tenant volume is high enough to amortize the managed-service premium versus reserved/self-hosted infrastructure a trade-off to revisit post-capstone.
- **No applicable tax incentive** unlike renewable-energy or energy-efficient hardware products, there is currently no Bangladeshi tax incentive scheme targeting AI/SaaS software specifically, so this factor (template item i) does not reduce ShopMate AI's cost structure; the closest analogue is the Bangladeshi ICT sector's income-tax exemption for software/IT-enabled service exporters, which could reduce corporate tax if the product is positioned for export revenue in a future commercial phase.

1.2 Effect of Resource Availability on Cost and Market Vulnerability

The “resource” ShopMate AI depends on is not a physical raw material but three external, third-party-controlled resources whose availability and pricing directly affect the product's cost and market position:

- **LLM/GPU compute capacity** frontier-model API availability and pricing are controlled by a small number of providers; a price increase or rate-limit tightening is the software-industry equivalent of a raw-material shortage and was already flagged as a cost risk in the CO1 proposal (Section 3.6).
- **Platform API access (Meta/WhatsApp)** the entire product depends on continued, policy-compliant access to Meta's Messenger/Instagram/WhatsApp Business Platform APIs; a change in Meta's API terms, pricing (e.g., WhatsApp's per-conversation pricing model), or approval requirements is a direct market-vulnerability factor outside the team's control.

- **Skilled local NLP/Bangla-language talent** as a comparatively scarce specialization, the cost of maintaining and improving Bangla/Banglish support over time is more labour-cost-sensitive than a purely English-language competitor's would be.

2. Environmental Impact of the Product

2.1 Emissions

ShopMate AI has no direct manufacturing emissions (template item i does not apply to a software product), but it has an indirect carbon footprint through the electricity consumed by (a) cloud compute for its own services and (b) third-party LLM inference calls. Two design choices reduce this footprint directly:

- **Tiered model routing** routing the large majority of AI calls (language ID, intent, complaint classification) to smaller, cheaper, and correspondingly less energy-intensive models, reserving the largest model for only the final customer-facing generation step, reduces average energy-per-message compared to a design that called a frontier model for every step.
- **Managed cloud infrastructure** using a major cloud provider's shared, high-utilization data centers is generally more energy-efficient per compute-unit than dedicated low-utilization on-premise servers would be, due to economies of scale in cooling and hardware utilization.

2.2 Change in Consumption / Use Patterns

The product's primary environmental effect is indirect, through the commerce behavior it automates rather than through the software itself:

- **Positive: reduced redundant/duplicate deliveries** faster, catalog-accurate responses (the grounding guardrail from CO2, Section 3.2) reduce order errors that would otherwise cause return shipments each avoided return-and-reship cycle avoids the associated fuel consumption and packaging waste.
- **Positive: paperless commerce** moving Tier-1 sales conversations and order confirmation out of manual/paper-based record keeping (common among the target f-commerce merchant segment) into a digital record reduces paper consumption at the merchant level.
- **Negative / neutral: increased order volume** if the AI genuinely increases conversion (the CO1 proposal's “order uplift” success metric, Section 3.5), the net physical-goods delivery volume for a given merchant may increase, which is a positive business outcome but a mild negative from a pure emissions-per-merchant perspective; this trade-off is inherent to any tool that increases e-commerce sales.

2.3 Reliance on Scarce vs. Abundant Resources

The system's primary “resource” dependency is GPU/accelerator hardware (used by the LLM providers ShopMate AI calls, not owned by the project itself). GPU-class chips currently rely on advanced semiconductor fabrication and a concentrated global supply chain, making large-model inference availability more supply-constrained than commodity CPU compute. This is mitigated at the architecture level by the tiered-routing design, which keeps ShopMate AI's own demand

weighted toward smaller, more widely available models rather than exclusively the most GPU-intensive frontier models.

3. Social Impact of the Product

3.1 Impact on People's Lives

ShopMate AI is designed to have a primarily positive social impact on two groups:

- **Merchants** small Bangladeshi e-commerce sellers, often solo operators or very small teams, are freed from round-the-clock manual message-answering, which the CO1 proposal identifies as directly consuming disproportionate staff time (Section 1.2); this can improve working hours, reduce burnout, and let a small business owner compete with larger, better-staffed competitors.
- **Customers** faster, more accurate responses (grounded in real catalog data rather than a possibly-distracted human agent) and native Bangla/Banglish support improve the shopping experience for a customer segment that mainstream English-first chatbot tools underserve.

A potential negative effect to acknowledge honestly: automating Tier-1 responses could reduce demand for entry-level customer-service labor at merchants who currently employ dedicated reply staff. The system is deliberately scoped to augment rather than fully replace merchant staff all AI drafts in the Unified Inbox require a human “Send Now” action by default (CO2, Section 2), and complaints are always routed to a human rather than auto-resolved which preserves a human role in the highest-judgment interactions even as routine Tier-1 volume is automated.

3.2 Community and Personal Needs Addressed

- Enables very small, resource-constrained merchants to offer service quality (24/7 responsiveness, consistent accuracy) previously only affordable to larger businesses with dedicated support teams.
- Addresses a specifically local need native Bangla/Banglish conversational support that is underserved by the mainstream, English-first chatbot tooling reviewed in the CO1 literature review (Section 2.2).

3.3 Safety and Health Concerns

- **Financial/consumer-protection safety** the hard grounding rule (never state an ungrounded price, stock level, or delivery promise; CO2, Section 3.2–3.3) exists specifically to prevent the AI from causing real financial harm to a customer through a hallucinated commitment a merchant cannot honor.
- **Data-handling safety** the system processes real customer personal data (names, phone numbers, order/delivery details); the CO1 proposal's risk analysis (Section 3.6) already commits to using only seeded synthetic data during development and evaluation, never live customer PII, specifically to avoid a data-safety incident during the capstone.
- **No physical/bodily-health hazard** as a pure software product with no hardware component, ShopMate AI presents no direct physical safety or occupational-health risk (template items

on physical safety are not applicable beyond the digital-financial and data-privacy dimensions addressed above).

3.4 Regulatory Constraints Addressing Social/Environmental Concerns

These overlap substantially with the legal considerations in Section 4 (data protection law, consumer protection law, and platform policy) and are cross-referenced there rather than duplicated.

4. Legal Considerations and Constraints

As a software SaaS product, ShopMate AI is not subject to hardware-oriented standards such as electric code, powerline communication, or GPIB/PCI/PXI bus standards (template items vi, vii, and xi are not applicable); the standards and regulations that materially constrain its design are summarized below.

Category	Relevant Standard / Regulation	Design Impact
Data protection	Bangladesh Data Protection Act (draft/in force per latest government notice); analogous principles from GDPR are followed as a best-practice baseline given many South/Southeast Asian markets are moving toward GDPR-aligned frameworks.	Multi-tenant row-level data isolation (CO2, Section 3.3/4.3); use of only seeded synthetic data in development and demos (CO1, Section 3.6); a defined data-retention policy for message/PII data to be finalized before any real customer data is processed.
Platform policy	Meta Platform Terms, WhatsApp Business Messaging Policy, WhatsApp Business Platform 24-hour customer-service messaging window.	The Delivery Worker checks each conversation's last-customer-message timestamp before every send and falls back to an approved template message once the window closes (CO2, Section 3.3); Integrations Hub UI proactively surfaces token-expiry/re-authentication needs (CO2, Section 2.1).
Payment / financial	PCI-DSS (Payment Card Industry Data Security Standard) for any checkout/payment-link flow.	The system never stores raw card data itself; checkout is generated as a secure payment link and processed by a PCI-DSS-compliant third-party payment gateway rather than in-house (CO1, Section 4.1 architecture).
Consumer protection	Bangladesh Consumer Rights Protection Act, 2009 (misleading claims / false advertising provisions relevant to AI-quoted prices and promises).	Directly motivates the hard grounding rule (CO2, Section 3.2–3.3) the AI must not state a price, stock level, or delivery promise it cannot substantiate from the merchant's own catalog data.
Information security	ISO/IEC 27001 information-security-management principles (used as a design reference, not a certification claim at capstone stage); OWASP API Security Top 10 for the webhook/ingestion surface.	Signature-verified, deduplicated webhook ingestion (CO2, Section 3.2); per-tenant rate limiting (CO2, Section 6.4); encryption in transit via TLS across all channel and internal API traffic.

Cryptographic / transport standards	TLS 1.2+/1.3 for data in transit; AES-256 for data at rest (as already advertised on the Sign-Up screen, CO2 Figure/Section 2.1).	All external webhook and dashboard traffic is served over HTTPS/TLS; database and object-storage encryption at rest is enabled by default on the managed cloud services used.
Authentication / authorization standards	OAuth 2.0 (used for Facebook/Instagram/WhatsApp channel connection and Google/Facebook social sign-in on the login screen, CO2 Section 2.1); role-based access control for dashboard users.	Channel connection flows in the Integrations Hub rely on each platform's OAuth 2.0 authorization flow rather than storing raw platform passwords; dashboard user roles (owner/agent/admin) gate access to billing, catalog editing, and persona configuration.
Software engineering / API standards	REST/HTTP API conventions; JSON Schema for the unified message contract (CO1, Section 4.3); semantic versioning for internal service APIs.	The channel-agnostic “Unified Message Contract” that every adapter normalizes into (CO1 proposal, Section 4.3 critical path item 1) is itself a lightweight internal standard that lets services and adapters evolve independently.
Software licensing	Open-source license compliance for third-party libraries used (e.g., MIT/Apache-2.0-licensed packages in the Node.js/Python stack).	All third-party dependencies are reviewed for license compatibility before inclusion; no GPL-licensed component is embedded in a way that would trigger copyleft obligations on the proprietary codebase.
